

Test and Reliability Exercise Sheet 7

Exercise 1

Your task is to develop an SBST for the simplified SPI master in figure 1. The SPI master is connected to a RISC-V CPU and is accessible via memory mapped IO (MMIO). A register documentation for the SPI master is given in table 1. Transform the patterns given in table 2 into C/C++ code and evaluate the SBST's coverage:

- a) Download and extract the template for the SBST on Linux / WSL, and open a shell in the directory. For distributions other than Ubuntu 24.04 LTS, use an Ubuntu 24.04 LTS Docker container via:

```
make docker
```

Download the PicoRV32 RISC-V core and install required dependencies via:

```
sudo make install_dependencies  
make vendor_processor # PicoRV32 RISC-V processor core
```

Download and extract the pre-built tools or build them manually via:

```
make build_verilator # Open-source circuit simulator  
make build_gcc # RISC-V GCC compiler toolchain
```

- b) Transfer the patterns into register read and write transactions in the firmware code. Modify the main function of `src_injection/firmware.cpp` and add the respective C/C++ code. Use the function `sbst_exit` to signal the detection of a fault by passing a result $\neq 0$.

```
void main() {  
    spi0.config = 0;  
    if (spi0.config != 0) {  
        sbst_exit(1); // A fault was detected  
    } else {  
        sbst_exit(0); // No fault was detected  
    }  
}
```

- c) Evaluate the SBST fault coverage on the gate-level version of the SPI master via:

```
make out_gate_firmware out_gate_verilator  
make out_gate_run
```

What is the resulting coverage of the patterns given?

- d) Try to improve the coverage of the SBST by adding and tinkering with tests.

Deadline: Discussion offline in the Forum, Solution will be provided on July 23rd.

Table 1: Registers of SPI master

Register	Read/Write	Description
STATUS[0]	Read-Only	0 if SPI is idle, 1 if SPI is busy
CONFIG[0]	Read-Write	0 for normal mode, 1 for loopback / test mode (connects MOSI with MISO internally to read back written data)
SEND[18:16]	Write-Only	SPI device to be selected via SS output (0 to 7)
SEND[15:0]	Write-Only	16 bits of data to be sent to the device
RECEIVE[18:16]	Read-Only	SPI device that was selected during the last transmission (0 to 7)
RECEIVE[15:0]	Read-Only	16 bits of data that was read from the device

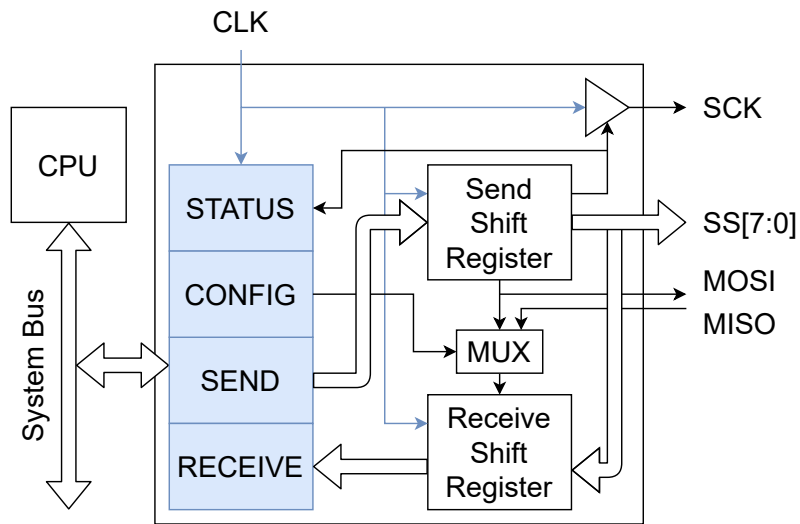


Figure 1: Multidrop SPI master (CPOL=1, CPHA=1) with loopback / test mode

Table 2: Patterns for circuit under test

Pattern	Timeframe	CONFIG	STATUS	SEND	RECEIVE
1	0	W 1 ₂	—	—	—
	1	R 1 ₂	—	—	—
2	0	W 1 ₂	—	—	—
	1	—	—	W 00000 ₁₆	—
	2	—	—	—	R 00000 ₁₆
3	0	W 1 ₂	—	—	—
	1	—	—	W 1FFFF ₁₆	—
	2	—	—	—	R 1FFFF ₁₆
4	0	W 1 ₂	—	—	—
	1	—	—	W 00000 ₁₆	—
	2	—	R 1 ₂	—	—